

In [1]: **import** numpy **as** np

#Seasons

```
Seasons = ["2010", "2011", "2012", "2013", "2014", "2015", "2016", "2017", "2018", "2019"]
Sdict = {"2010":0, "2011":1, "2012":2, "2013":3, "2014":4, "2015":5, "2016":6, "2017":7
```

#Players

```
Players = ["Sachin", "Rahul", "Smith", "Sami", "Pollard", "Morris", "Samson", "Dhoni", "Kohli", "Sky"]
Pdict = {"Sachin":0, "Rahul":1, "Smith":2, "Sami":3, "Pollard":4, "Morris":5, "Samson":6, "Dhoni":7, "Kohli":8, "Sky":9}
```

#Salaries

```
Sachin_Salary = [15946875, 17718750, 19490625, 21262500, 23034375, 24806250, 25244493, 26000000, 27000000, 28000000]
Rahul_Salary = [12000000, 12744189, 13488377, 14232567, 14976754, 16324500, 18038573, 19000000, 20000000, 21000000]
Smith_Salary = [4621800, 5828090, 13041250, 14410581, 15779912, 14500000, 16022500, 17000000, 18000000, 19000000]
Sami_Salary = [3713640, 4694041, 13041250, 14410581, 15779912, 17149243, 18518574, 19450000, 20000000, 21000000]
Pollard_Salary = [4493160, 4806720, 6061274, 13758000, 15202590, 16647180, 18091770, 19000000, 20000000, 21000000]
Morris_Salary = [3348000, 4235220, 12455000, 14410581, 15779912, 14500000, 16022500, 17000000, 18000000, 19000000]
Samson_Salary = [3144240, 3380160, 3615960, 4574189, 13520500, 14940153, 16359805, 17770000, 18000000, 19000000]
Dhoni_Salary = [0, 0, 4171200, 4484040, 4796880, 6053663, 15506632, 16669630, 17832627, 19000000]
Kohli_Salary = [0, 0, 0, 4822800, 5184480, 5546160, 6993708, 16402500, 17632688, 18862875]
Sky_Salary = [3031920, 3841443, 13041250, 14410581, 15779912, 14200000, 15691000, 17182000, 18000000, 19000000]
```

#Matrix

```
Salary = np.array([Sachin_Salary, Rahul_Salary, Smith_Salary, Sami_Salary, Pollard_Salary, Morris_Salary, Samson_Salary, Dhoni_Salary, Kohli_Salary, Sky_Salary])
```

#Games

```
Sachin_G = [80, 77, 82, 82, 73, 82, 58, 78, 6, 35]
Rahul_G = [82, 57, 82, 79, 76, 72, 60, 72, 79, 80]
Smith_G = [79, 78, 75, 81, 76, 79, 62, 76, 77, 69]
Sami_G = [80, 65, 77, 66, 69, 77, 55, 67, 77, 40]
Pollard_G = [82, 82, 82, 79, 82, 78, 54, 76, 71, 41]
Morris_G = [70, 69, 67, 77, 70, 77, 57, 74, 79, 44]
Samson_G = [78, 64, 80, 78, 45, 80, 60, 70, 62, 82]
Dhoni_G = [35, 35, 80, 74, 82, 78, 66, 81, 81, 27]
Kohli_G = [40, 40, 40, 81, 78, 81, 39, 0, 10, 51]
Sky_G = [75, 51, 51, 79, 77, 76, 49, 69, 54, 62]
```

#Matrix

```
Games = np.array([Sachin_G, Rahul_G, Smith_G, Sami_G, Pollard_G, Morris_G, Samson_G, Dhoni_G, Kohli_G, Sky_G])
```

#Points

```
Sachin_PTS = [2832, 2430, 2323, 2201, 1970, 2078, 1616, 2133, 83, 782]
Rahul_PTS = [1653, 1426, 1779, 1688, 1619, 1312, 1129, 1170, 1245, 1154]
Smith_PTS = [2478, 2132, 2250, 2304, 2258, 2111, 1683, 2036, 2089, 1743]
Sami_PTS = [2122, 1881, 1978, 1504, 1943, 1970, 1245, 1920, 2112, 966]
Pollard_PTS = [1292, 1443, 1695, 1624, 1503, 1784, 1113, 1296, 1297, 646]
Morris_PTS = [1572, 1561, 1496, 1746, 1678, 1438, 1025, 1232, 1281, 928]
Samson_PTS = [1258, 1104, 1684, 1781, 841, 1268, 1189, 1186, 1185, 1564]
Dhoni_PTS = [903, 903, 1624, 1871, 2472, 2161, 1850, 2280, 2593, 686]
Kohli_PTS = [597, 597, 597, 1361, 1619, 2026, 852, 0, 159, 904]
Sky_PTS = [2040, 1397, 1254, 2386, 2045, 1941, 1082, 1463, 1028, 1331]
```

#Matrix

```
Points = np.array([Sachin_PTS, Rahul_PTS, Smith_PTS, Sami_PTS, Pollard_PTS, Morris_PTS, Samson_PTS, Dhoni_PTS, Kohli_PTS, Sky_PTS])
```

```
In [2]: import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
import warnings
warnings.filterwarnings('ignore')
```

```
In [3]: Players
```

```
Out[3]: ['Sachin',
        'Rahul',
        'Smith',
        'Sami',
        'Pollard',
        'Morris',
        'Samson',
        'Dhoni',
        'Kohli',
        'Sky']
```

```
In [4]: print(Games)
```

```
[[80 77 82 82 73 82 58 78  6 35]
 [82 57 82 79 76 72 60 72 79 80]
 [79 78 75 81 76 79 62 76 77 69]
 [80 65 77 66 69 77 55 67 77 40]
 [82 82 82 79 82 78 54 76 71 41]
 [70 69 67 77 70 77 57 74 79 44]
 [78 64 80 78 45 80 60 70 62 82]
 [35 35 80 74 82 78 66 81 81 27]
 [40 40 40 81 78 81 39  0 10 51]
 [75 51 51 79 77 76 49 69 54 62]]
```

```
In [5]: print(Seasons)
```

```
['2010', '2011', '2012', '2013', '2014', '2015', '2016', '2017', '2018', '2019']
```

```
In [6]: print(Salary)
```

```
[[15946875 17718750 19490625 21262500 23034375 24806250 25244493 27849149
 30453805 23500000]
 [12000000 12744189 13488377 14232567 14976754 16324500 18038573 19752645
 21466718 23180790]
 [ 4621800  5828090 13041250 14410581 15779912 14500000 16022500 17545000
 19067500 20644400]
 [ 3713640  4694041 13041250 14410581 15779912 17149243 18518574 19450000
 22407474 22458000]
 [ 4493160  4806720  6061274 13758000 15202590 16647180 18091770 19536360
 20513178 21436271]
 [ 3348000  4235220 12455000 14410581 15779912 14500000 16022500 17545000
 19067500 20644400]
 [ 3144240  3380160  3615960  4574189 13520500 14940153 16359805 17779458
 18668431 20068563]
 [      0      0 4171200  4484040  4796880  6053663 15506632 16669630
 17832627 18995624]
 [      0      0      0 4822800  5184480  5546160  6993708 16402500
 17632688 18862875]
 [ 3031920  3841443 13041250 14410581 15779912 14200000 15691000 17182000
 18673000 15000000]]
```

```
In [7]: print(Salary//Games)
```

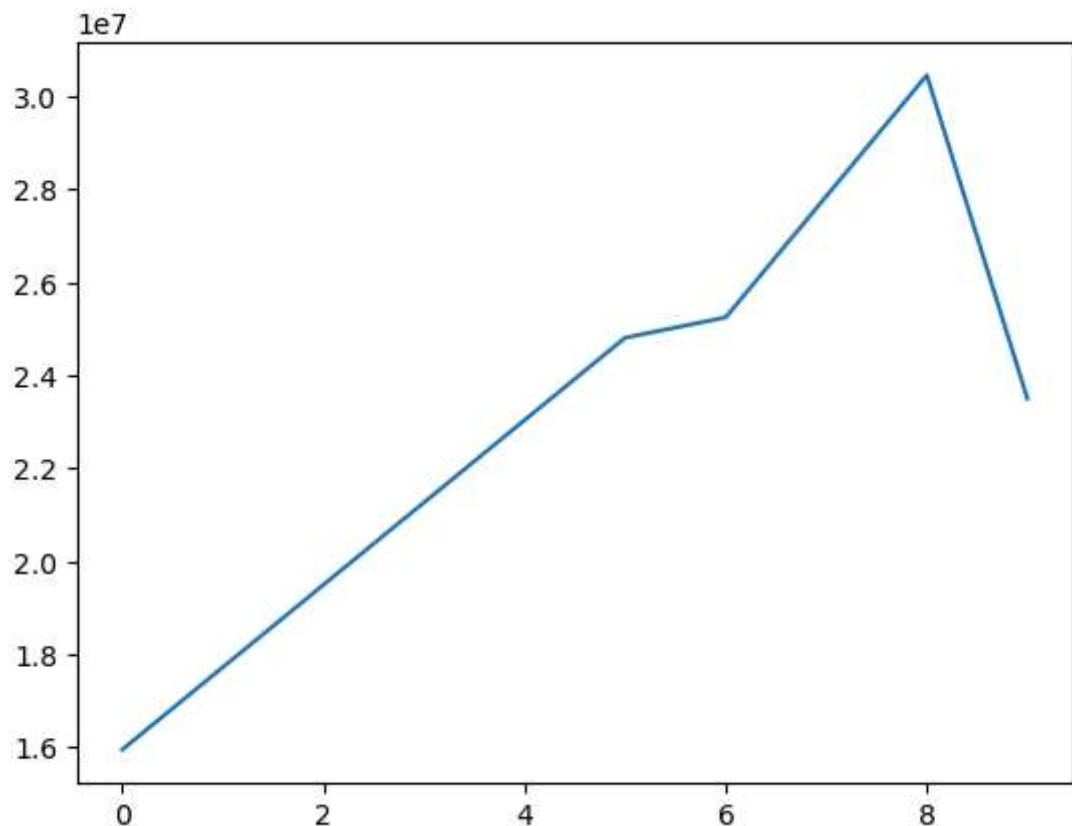
```
[[ 199335  230113  237690  259298  315539  302515  435249  357040  5075634
   671428]
 [ 146341  223582  164492  180159  197062  226729  300642  274342  271730
   289759]
 [  58503   74719  173883  177908  207630  183544  258427  230855  247629
   299194]
 [  46420   72216  169366  218342  228694  222717  336701  290298  291006
   561450]
 [  54794   58618   73917  174151  185397  213425  335032  257057  288918
   522835]
 [  47828   61380  185895  187150  225427  188311  281096  237094  241360
   469190]
 [  40310   52815   45199   58643  300455  186751  272663  253992  301103
   244738]
 [      0      0  52140   60595   58498   77611  234948  205797  220155
   703541]
 [      0      0      0  59540   66467   68471  179325      0 1763268
   369860]
 [  40425   75322  255710  182412  204933  186842  320224  249014  345796
   241935]]
```

```
In [8]: Salary[0]
```

```
Out[8]: array([15946875, 17718750, 19490625, 21262500, 23034375, 24806250,
              25244493, 27849149, 30453805, 23500000])
```

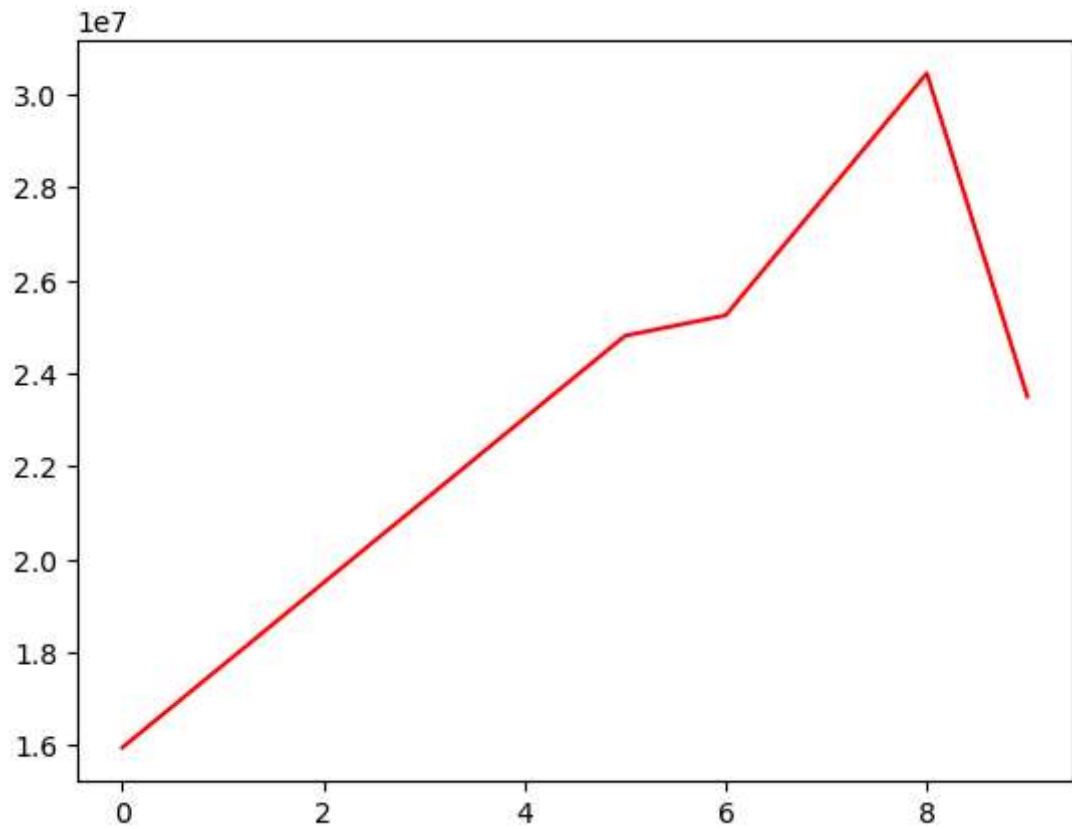
```
In [9]: plt.plot(Salary[0])
```

```
Out[9]: [<matplotlib.lines.Line2D at 0x1be6750ed50>]
```



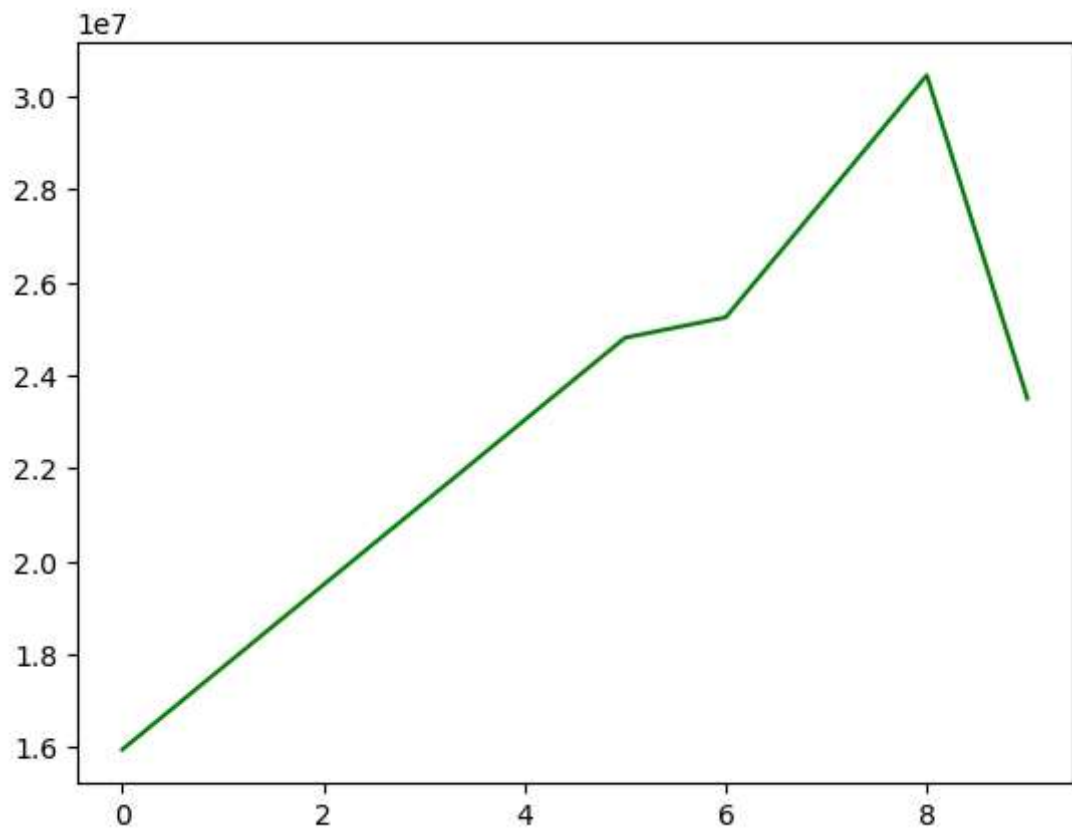
```
In [10]: plt.plot(Salary[0],c='r')
```

```
Out[10]: [<matplotlib.lines.Line2D at 0x1be676cb4d0>]
```



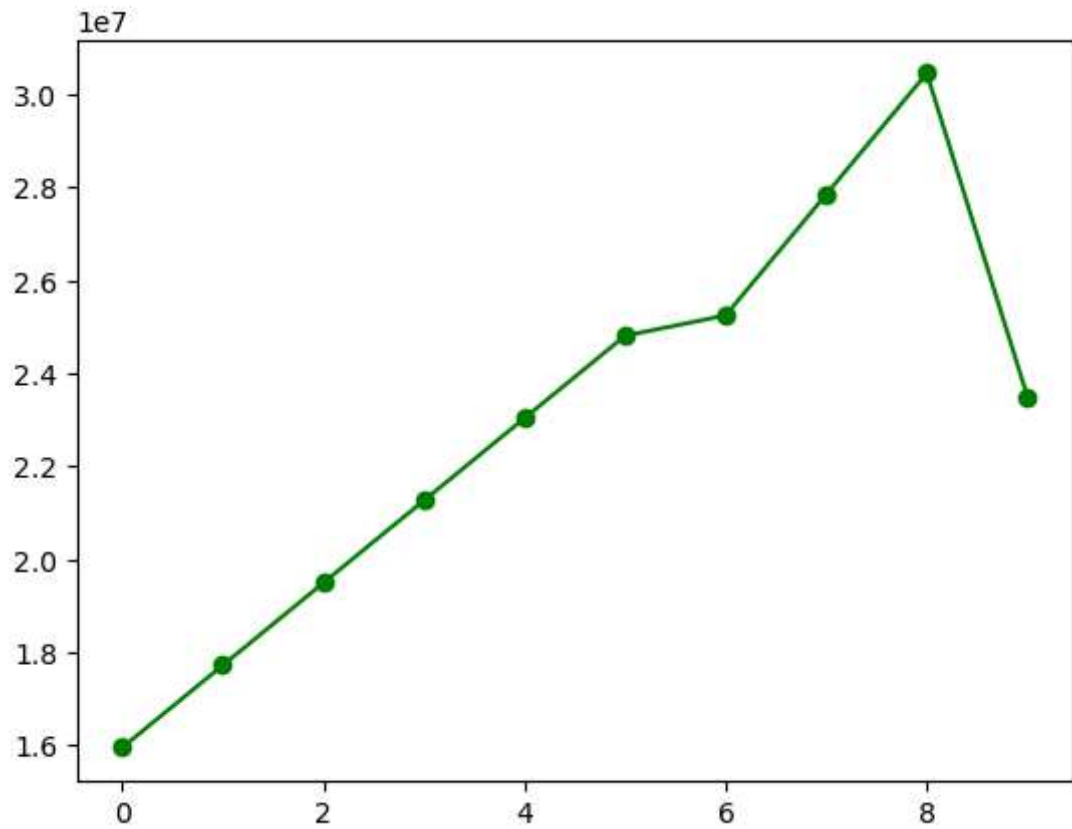
```
In [11]: plt.plot(Salary[0],c='g')
```

```
Out[11]: [<matplotlib.lines.Line2D at 0x1be6789a490>]
```



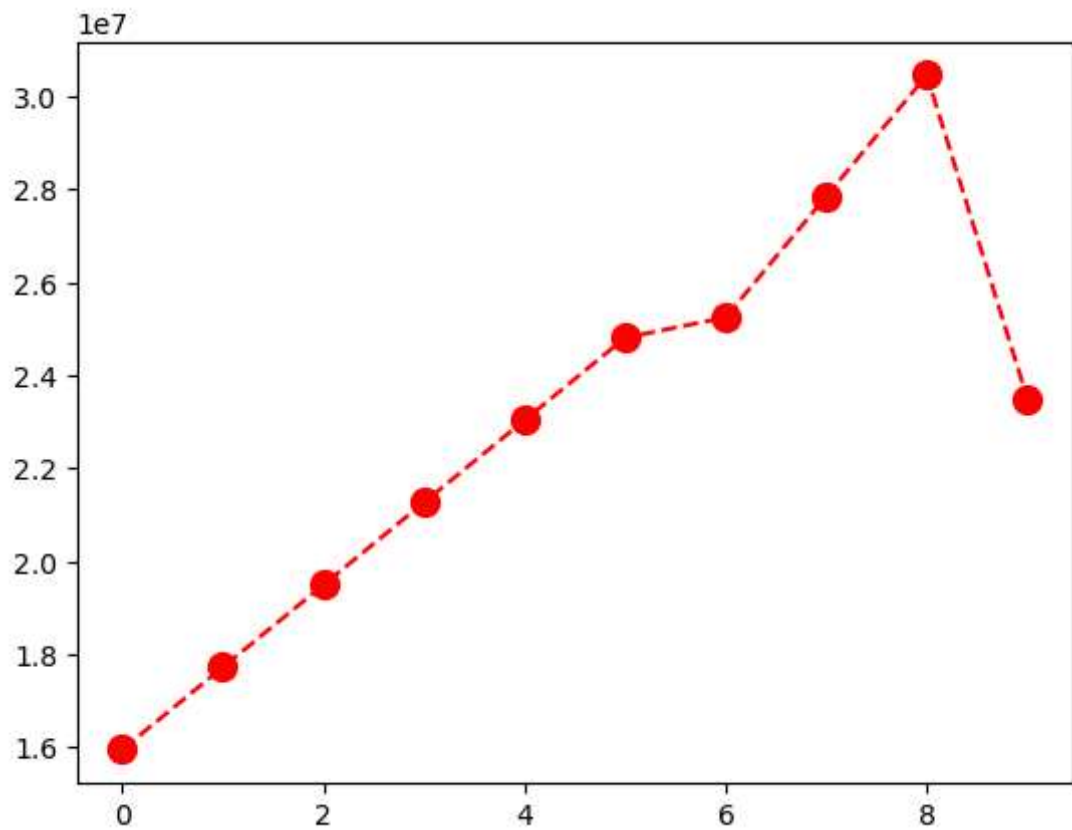
```
In [12]: plt.plot(Salary[0],c='g',marker='o')
```

```
Out[12]: [<matplotlib.lines.Line2D at 0x1be67914a50>]
```



```
In [13]: plt.plot(Salary[0],c='r',marker='o',ms=10,ls='--')
```

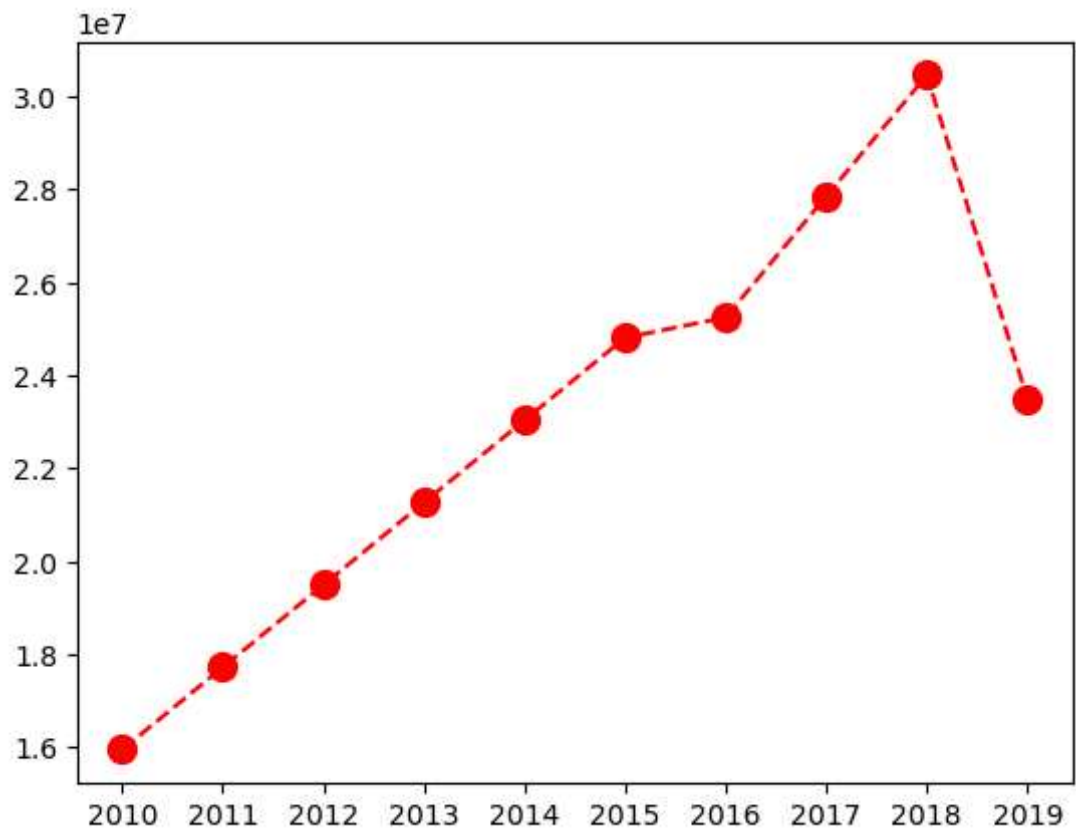
```
Out[13]: [<matplotlib.lines.Line2D at 0x1be6797efd0>]
```



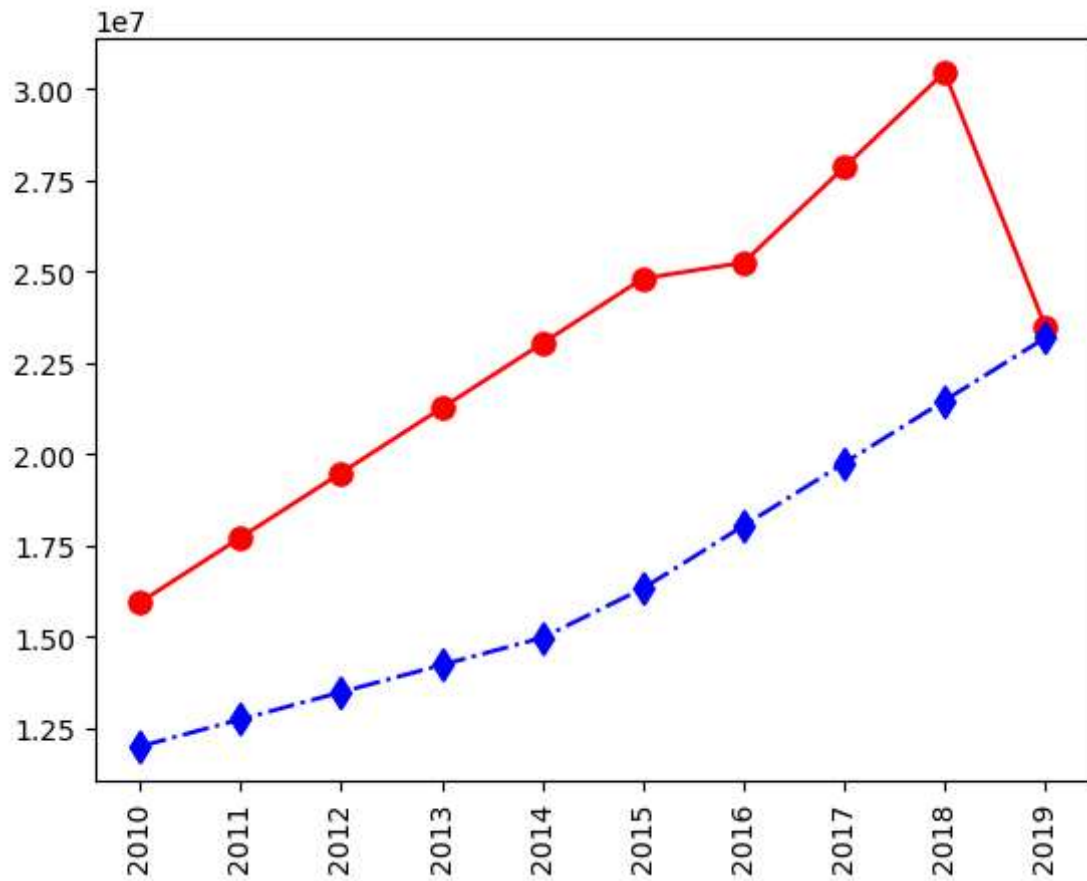
```
In [14]: Sdict
```

```
Out[14]: {'2010': 0,  
          '2011': 1,  
          '2012': 2,  
          '2013': 3,  
          '2014': 4,  
          '2015': 5,  
          '2016': 6,  
          '2017': 7,  
          '2018': 8,  
          '2019': 9}
```

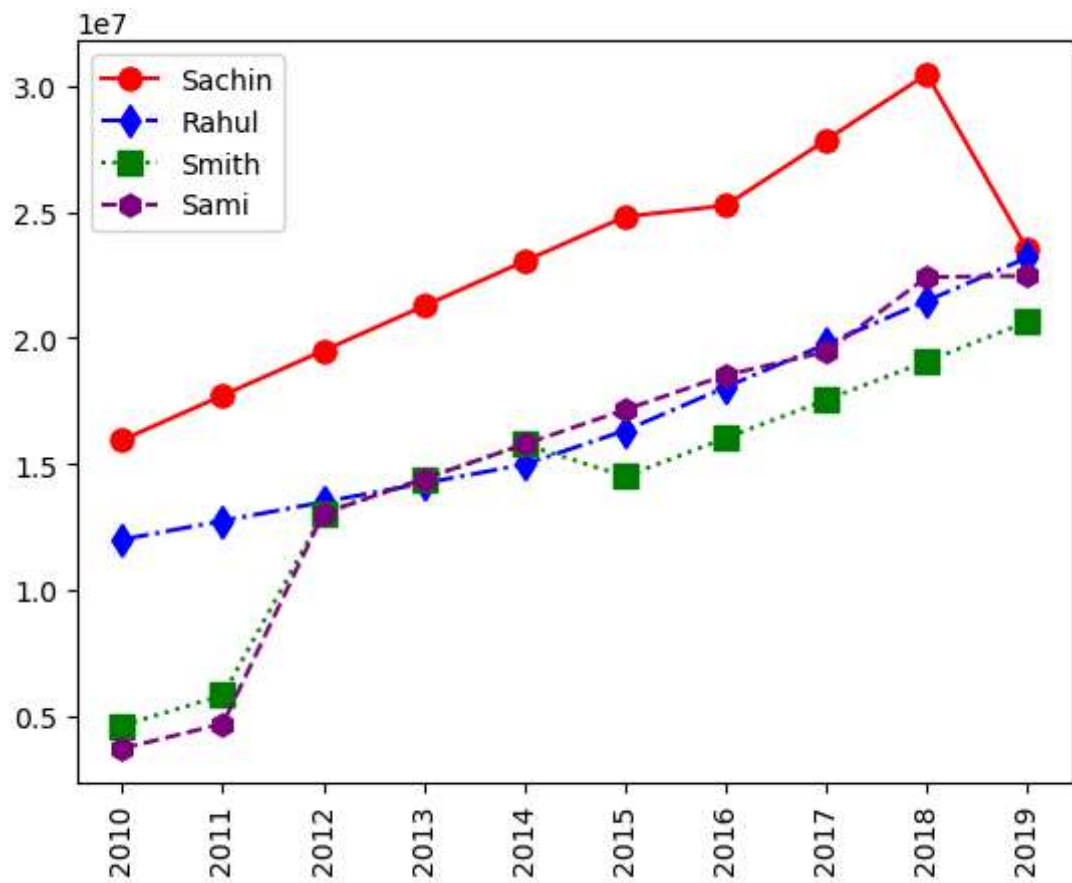
```
In [15]: plt.plot(Salary[0],c='r',marker='o',ms=10,ls='--')  
plt.xticks(list(range(0,10)),Seasons)  
plt.show()
```



```
In [37]: plt.plot(Salary[0],c='r',marker='o',ms=8,ls='-',label=Players[0])  
plt.plot(Salary[1],c='b',marker='d',ms=8,ls='-.',label=Players[1])  
  
plt.xticks(list(range(0,10)),Seasons, rotation = 'vertical')  
  
plt.show()
```



```
In [35]: plt.plot(Salary[0],c='r',marker='o',ms=8,ls='-',label=Players[0])
plt.plot(Salary[1],c='b',marker='d',ms=8,ls='-',label=Players[1])
plt.plot(Salary[2],c='g',marker='s',ms=8,ls=':',label=Players[2])
plt.plot(Salary[3],c='purple',marker='h',ms=8,ls='--',label=Players[3])
plt.xticks(list(range(0,10)),Seasons, rotation = 'vertical')
plt.legend(loc = 'upper left',bbox_to_anchor=(0.5,1))
plt.legend()
plt.show()
```



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